Republic of the Philippines

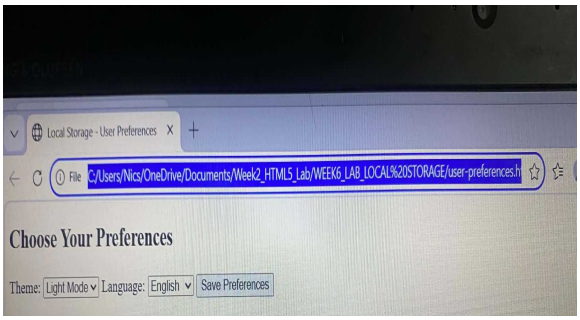
**Bicol University Polangui**

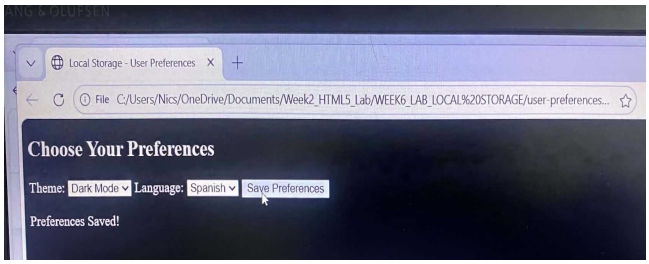
Computer Studies Department

A. Y 2026-2027

# CLIENT-SIDE STORAGE & API INTEGRATION WITH JAVASCRIPT

## STEP 1: Working with Local Storage



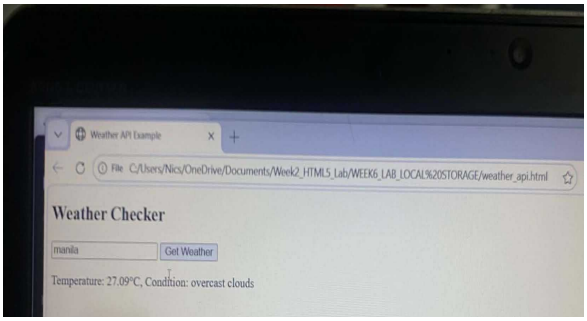



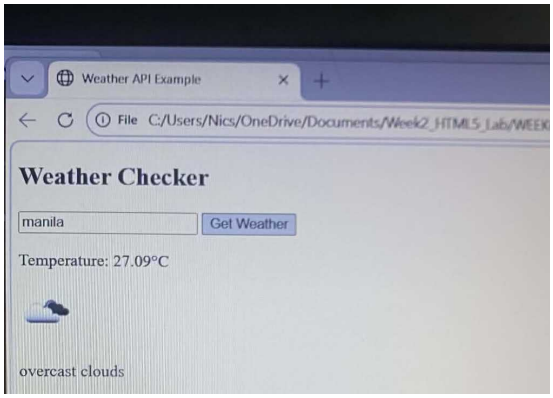

**Description:**In this activity, I learned how to use **local storage** to save data directly in the user's browser. This allows information to remain available even after the webpage is refreshed or closed. I practiced storing, retrieving, and removing data using Bootstrap.

## STEP 1.1: Enhanced Local Storage – Saving User Preferences

In this step, I enhanced the local storage activity by saving **user preferences**, such as selected settings or choices. The saved preferences can be automatically retrieved when the user returns to the webpage, creating a more personalized and convenient experience.

## STEP 2:Fetching Real-Time Weather Data



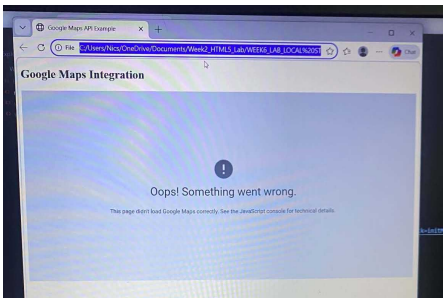

**Description:** In this activity, I learned how to retrieve **real-time weather information** from an external service. Using Bootstrap and API requests, the application can receive current weather data and display it dynamically on the webpage.

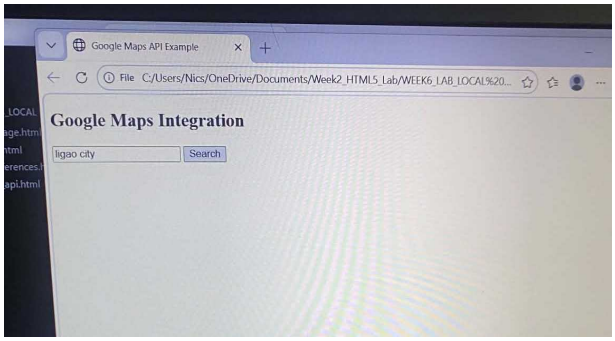
## Step 2.1: Fetching Weather Data Using the OpenWeatherMap API

For this step, I used the **OpenWeatherMap API** to obtain weather information based on a specific location. I learned how to send a request to an API, process the returned JSON data, and display information such as temperature, weather conditions, and location on the webpage.

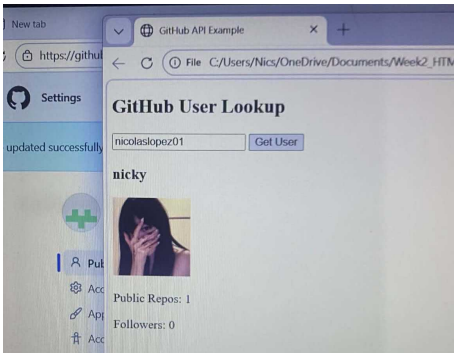
**Step 3 & 3.1: Displaying Google Maps With User Input**

In this step, I do not have a key card so I cannot open the google map but I copy the step even if I do not have the card to open the google for the code or link.




**Step 4:etching Social Media Data Using GitHub API**




**Description:**In this activity, I learned how to retrieve data from the **GitHub API**. I practiced sending API requests and displaying information from GitHub, such as user or repository data. This helped me understand how web applications can communicate with external platforms and use their available data.

**REFLECTION:**Through these activities, I learned how different web technologies can be combined to create more interactive and dynamic applications. Working with **local storage** taught me how to save information on the client side and maintain user preferences. I also learned how **APIs** allow a web page to communicate with external services and retrieve real-time information. The weather activity helped me understand how to make API requests and process JSON responses, while the Google Maps activity showed me how external mapping services can be integrated into a web application.

**REFERENCES:**

Google. (n.d.). Maps JavaScript API overview. Google for Developers.

GitHub. (n.d.). REST API documentation. GitHub Docs.